

SMRITI SINGH

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EDUCATION

Vellore Institute of Technology

June 2023 – Present

Bachelor of Technology in Computer Science Engineering (AI & ML)

CGPA: **9.22/10**

EXPERIENCE

Tata Steel

June 2026 – Present

AI/ML Intern

Jamshedpur, India

- Developed an end-to-end machine learning regression pipeline to predict molten steel temperature drop using historical production data.
- Processed and validated **3,287+** industrial records through data cleaning, missing value handling, outlier removal, and domain-specific validation.
- Engineered **6** domain-specific features and optimized a **Random Forest Regressor** using **RandomizedSearchCV**, achieving an MAE of **6.9°C**.
- Built a modular prediction system with preprocessing, training, batch inference, and a **Streamlit** web application for real-time temperature prediction following clean software engineering practices.

PROJECTS

MediMind AI – Intelligent Clinical Decision Support System

Jul 2026–Present

Live Demo — GitHub

Python, FastAPI, React, ChromaDB, Gemini, RAG

- Developed an AI-powered clinical decision support system using FastAPI, React.js, ChromaDB, and Google Gemini for intelligent medical report analysis.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB and Sentence Transformers for semantic search and evidence-grounded responses.
- Built scalable REST APIs and an automated PDF processing pipeline to extract laboratory parameters from CBC and LFT reports.
- Integrated Google Gemini with prompt engineering to generate clinically relevant summaries while reducing hallucinations.
- Developed an interactive React.js interface for report upload and AI-powered report interpretation.

Multi-Agent AI Research Assistant

Jun 2026–Jul 2026

Live Demo — GitHub

Python, LangChain, Gemini, Streamlit

- Designed and deployed a production-ready multi-agent AI system for automated web research.
- Developed custom tools integrating Tavily Search API and BeautifulSoup for information retrieval.
- Implemented Search, Writer, and Critic agents using prompt engineering and tool orchestration.
- Deployed on Render with an interactive Streamlit frontend.

Customer Churn Prediction System

May 2026–June 2026

Live Demo — GitHub

Python, Flask, Docker, Streamlit

- Developed and deployed a production-ready churn prediction platform using Flask, Docker, and Streamlit.
- Built an end-to-end ML pipeline for preprocessing, training, evaluation, and inference.
- Achieved **82% accuracy** and **0.86 ROC-AUC** using Logistic Regression.
- Designed an interactive dashboard for churn prediction and retention insights.

TECHNICAL SKILLS

Languages	Python, C++
Technologies	FastAPI, Flask, React, Streamlit, LangChain, RAG, LLMs, Scikit-learn, TensorFlow, PyTorch
Databases	MySQL, MongoDB, SQL
Core CS	DSA, OOP, DBMS, Operating Systems

CO-CURRICULAR ACTIVITIES

- **Co-Lead**, Design Team, IoT Club, VIT Bhopal – Led design initiatives and collaborated on organizing technical events.
- Won **two Silver Medals** in the **Bench Press** and **Squat** events at the **VIT Sports Fest**.

CERTIFICATIONS

- **Applied Machine Learning in Python** – University of Michigan (Coursera)
- **Machine Learning** in collaboration with Google for Developers